

Carnegie Mellon University · Heinz College

95–703 C: Database Management

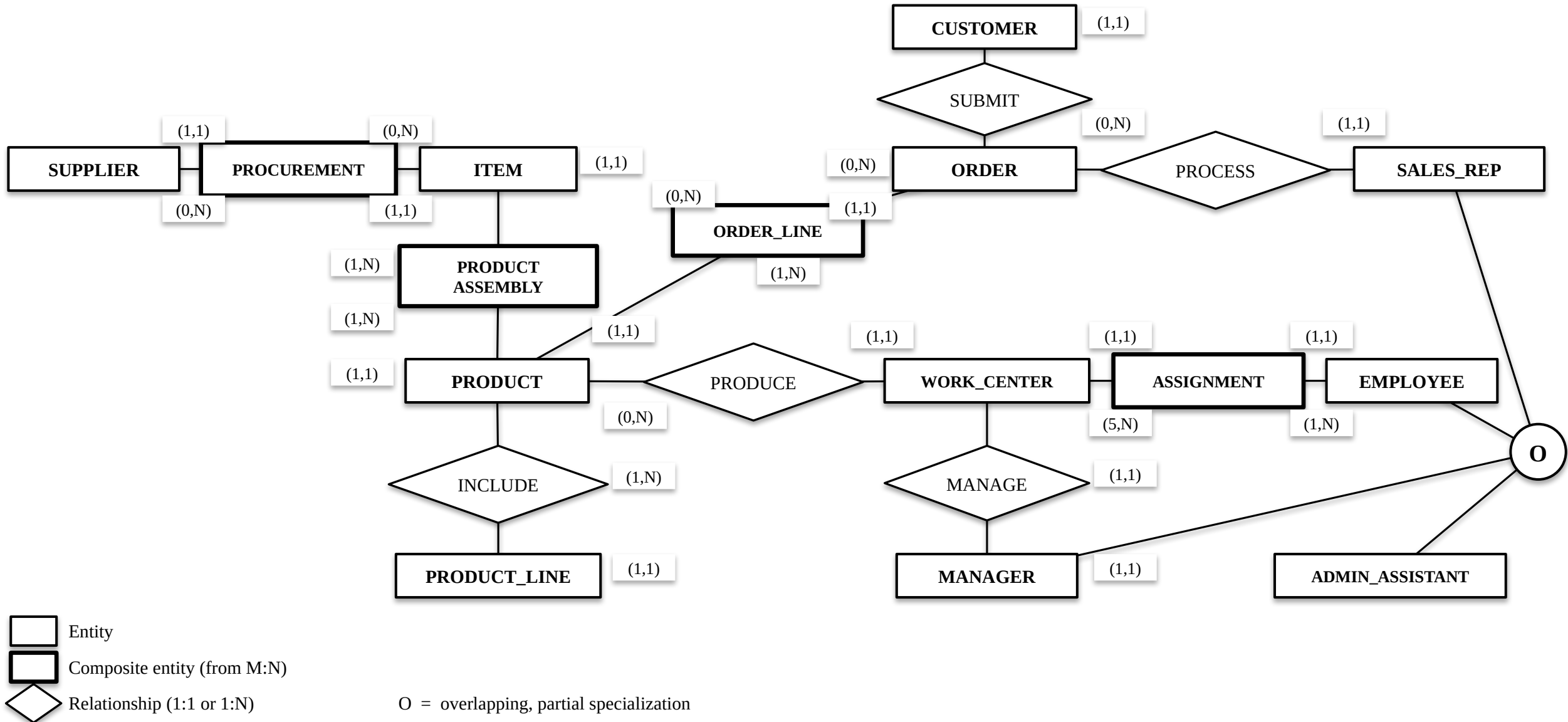
Homework #2 — Standard Solution (TA Answer Key)

Fall 2026

1. Conceptual model: ER diagram without attributes
2. Many-to-many relationships transformed into composite entities
3. Logical model: relational schema (PK underlined, FK marked with @)

Use this deck as the in-class key. Cardinality uses (min, max) look-across, matching the Painter-model template.

1. ER Diagram (no attributes; M:N → composite entities)



2. Relational schema (PK underlined; FK marked with @)

Each relation maps one entity, composite entity, or subclass. Identifying FKs that are also part of the PK are both underlined and tagged @.

SUPPLIER: Supplier ID, Supplier_Name, Supplier_Address, Supplier_Phone

PROCUREMENT: Item ID@, Procurement Date, Supplier_ID@, Quantity, Unit_Price

ITEM: Item ID, Description, Average_Unit_Cost, Units_On_Hand, Reorder_Point

PRODUCT_ASSEMBLY: Product ID@, Item ID@, Quantity_Needed

PRODUCT: Product ID, Product_Name, Product_Line_ID@, Work_Center_ID@

PRODUCT_LINE: Product Line ID, Product_Line_Name

CUSTOMER: Customer ID, Customer_Name, Customer_Email, Customer_Address, Customer_Phone

ORDER: Order ID, Order_Date, Customer_ID@, Sales_Rep_ID@

ORDER_LINE: Order ID@, Product ID@, Quantity_Ordered, Unit_Price

WORK_CENTER: Work Center ID, Work_Center_Name, Manager_ID@

ASSIGNMENT: Employee ID@, Work Center ID@

EMPLOYEE: Employee ID, First_Name, Last_Name, Address, Salary, Title

MANAGER: Employee ID@, Driver_License, Budget_Amount

ADMIN_ASSISTANT: Employee ID@, Accounting_Training, MS_Office_Level

SALES_REP: Employee ID@, Sales_Area, Email

Notes: WORK_CENTER.Manager_ID@ is unique (1:1 with MANAGER). CUSTOMER.Customer_Email may be null until the first order. (5,N) on ASSIGNMENT is an ER constraint — not encoded by the relation keys alone.

3. Mapping decisions · what to accept · common errors

Composite entities (only M:N)

- PROCUREMENT — SUPPLIER ↔ ITEM
- PRODUCT_ASSEMBLY — ITEM ↔ PRODUCT
- ORDER_LINE — ORDER ↔ PRODUCT
- ASSIGNMENT — EMPLOYEE ↔ WORK_CENTER
- 1:1 and 1:N stay diamonds; FK goes into the N-side (or either side of 1:1).

Required cardinalities

- Supplier may supply 0..N items over time; an item at a given date has 1 supplier.
- Item (1,N) products; product (1,N) items; store Quantity_NEEDED.
- Customer (0,N) orders (potential customers); each order (1,1) customer.
- Order (1,N) products; product (0,N) orders; store qty + unit price on ORDER_LINE.
- Work center (0,N) products; each product (1,1) work center.
- Employee (1,N) centers; work center (5,N) employees.
- Each work center (1,1) manager; each manager (1,1) work center.

Schema details worth full credit

- PROCUREMENT PK = (Item_ID, Procurement_Date); Supplier_ID@ is NOT in the PK — this enforces “at that time, one supplier”.
- ORDER.Customer_ID@ and ORDER.Sales_Rep_ID@ both mandatory (every order has one customer and one sales rep).
- WORK_CENTER.Manager_ID@ references MANAGER and is unique (1:1).
- Subclasses: MANAGER / SALES_REP / ADMIN_ASSISTANT with Employee_ID@ as PK/FK (overlapping, partial ISA).
- Customer_Email lives on CUSTOMER (nullable for potential customers). Sales-rep Email is a subclass attribute.

Do not award full credit if ...

- ASSIGNMENT shows employee participation (0,N) — the spec says every employee is assigned to ≥ 1 center.
- M:N left as a single relationship (no composite entity / no associative table).
- PROCESS attached to EMPLOYEE rather than SALES_REP (“only sales reps can process orders”).
- Procurement PK = (Supplier_ID, Item_ID, Date) — allows two suppliers for the same item on the same date.
- Specialization drawn as total (every employee is a manager/rep/assistant) — it is partial.
- Attribute ovals on the ER — the assignment asks for ER without attributes.